THE FLAT-AT-PRIMES CONDITION IS PRESERVED BY BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Flatness is preserved by base change. Algebra, Lemma \ref{algebra-lemma-flat-base-change}, cited at proof lines 4251-4253. Makes the base changed localised module flat.
- Flatness is preserved by localisation. Algebra, Lemma \ref{algebra-lemma-flat-localization}, cited at proof lines 4251-4253. Carries flatness to the localisation at the prime of the base change.

Source reference. Original lemma 05LR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7245–7245. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 05LP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7243–7243. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05LQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7244–7244. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 644–644. Navigation pointer to a statement the

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Source reference. Original lemma 00HT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 655–655. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *In Situation 05LP let $R' \to R''$ be an R-algebra map. Let $I' \subset R'$ and $I'R'' \subset I'' \subset R''$ be ideals. If (05LQ) holds for (R', I'), then (05LQ) holds for (R'', I'').*

Proof. Assume (05LQ) holds for (R', I'). Let $I''S'' + JS'' \subset \mathfrak{q}''$ be a prime of S''. Let $\mathfrak{q}' \subset S'$ be the corresponding prime of S'. Then both $I'S' \subset \mathfrak{q}'$ and $JS' \subset \mathfrak{q}'$ because the corresponding conditions hold for $\mathfrak{q}''$. Note that $(M'')_{\mathfrak{q}''}$ is a localization of the base change $M'_{\mathfrak{q}'} \otimes_R R''$. Hence $(M'')_{\mathfrak{q}''}$ is flat over R'' as a localization of a flat module, see Algebra, Lemmas 00HI and 00HT. □